Sylvester: A Benchmark and Empirical Study of Lifecycle-Aware Entity Resolution on Open Financial Data

Sheng Chang Li
Western University
London, Ontario, Canada
sli3244@uwo.ca

Abstract

Risk, compliance, and reference-data teams must follow a legal entity through mergers, rebrands, spinoffs, and jurisdiction changes, where its name and identifiers change by design. Snapshot entity-resolution benchmarks do not pose this task, and the open registry that records such events (GLEIF) does so only as sparse, partly administrative successor links. We introduce Sylvester, a public benchmark and empirical study of *lifecycle-aware* entity resolution: 125,519 labeled pairs over 3.3M entities from GLEIF Level-1/2, SEC EDGAR, and Wikidata, with held-out time, event-type, and jurisdiction splits. Because successors change names while hard negatives are mined to be name-similar, name similarity is anti-predictive on the released pool, so we evaluate every method on three candidate pools: the released pool, an unrelated miner's pool, and retrieval over the full universe. Methods that exploit the construction (inverted name similarity, 64–70% Hits@1 on time/event; a name-feature gradient-boosted matcher, 0.73–0.84) collapse off-pool. Among name-only methods, a 22M-parameter cross-encoder transfers (57–61% Hits@1 on time and event), 24–35 pp ahead of the best bi-encoder and ahead of six zero-shot LLMs. Input matters as much as architecture: given the full registry record, a gradient-boosted matcher beats that cross-encoder on-pool by 18 pp, and a cross-encoder over the same record wins the fielded setting by 12 pp (both $p < 10^{-4}$). Even so, the best system reaches only 21–39% Hits@1 end to end, because first-stage retrieval (Recall@20 of 33–50%) misses most true successors: the bottleneck is finding the candidate, not scoring it.

CCS Concepts

• **Information systems** → **Entity resolution**; • **Computing methodologies** → *Machine learning*; • **Applied computing** → Economics.

Keywords

entity resolution, record linkage, financial entities, lifecycle events, GLEIF, benchmark, cross-encoder

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1 Introduction

Twitter Inc. became X Corp. Facebook became Meta. Sprint was absorbed into T-Mobile. Each pair shares almost nothing on the surface (different legal names and tickers, sometimes different jurisdictions and legal forms), yet describes one continuing economic entity. We ask: *given two records observed at different times, do they describe the same economic entity, accounting for any lifecycle event between them?*

Who needs this, and why it matters. Several groups do this linkage routinely. Risk teams aggregate counterparty exposure by legal entity, and a stale identifier splits one exposure in two. Compliance teams screen clients against sanctions and KYC records, where a renamed or absorbed entity must still match. Reference-data teams maintain security masters and corporate-action histories, and researchers need the same links to build firm-level panels free of survivorship bias. The costly error is the same in each case: one entity treated as two, or a successor never found, and pipelines that make it fail silently.

Why it is hard. Snapshot entity resolution (ER) relies on surface similarity and does it well: LLM matchers reach 99% F1 on point-in-time records [10]. Lifecycle events remove that signal. A successor often inherits an acquirer's name and can share no tokens with its predecessor (Figure 1), while the most confusable candidates are namesakes, siblings, and parents that share the name but are legally distinct. Events are skewed (absorption is 94.5% of our held-out-event test set), and in deployment the successor must first be retrieved from millions of entities.

Why no benchmark existed. Commercial reference-data vendors maintain lifecycle links, but their data is licensed and cannot be redistributed. The open GLEIF registry holds the raw pieces but not a benchmark. Its `SuccessorEntity` links are filed voluntarily, only 67% resolve to a registered successor, 5.4% of those are administrative re-registrations, and the event log was sparse before 2022 (§3). Its Level-2 relationship records encode ownership and fund hierarchy, not succession: only 0.1% of our positives have a Level-2 edge between predecessor and successor (§3.1). A benchmark also needs hard negatives, held-out splits, and a full candidate universe, which the registry does not provide, and existing ER benchmarks are snapshot-framed (§2).

This paper. We build that benchmark and use it for an empirical study. The models are standard on purpose: string baselines, gradient-boosted matchers, off-the-shelf sentence-encoder bi- and

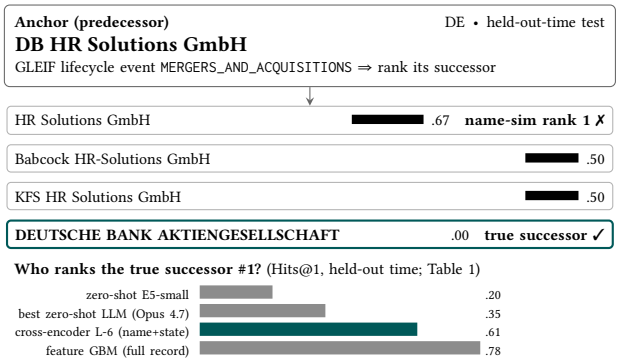


Figure 1: The task. ***Top:*** **a real held-out-time test anchor (4 of 7 candidates; bars = token Jaccard). Name similarity ranks a coincidental collision first and the true successor, with zero shared tokens, last (§6.1).** ***Bottom:*** **released-pool Hits@1 for four methods (a name-only exploit reaches .77 here but collapses off-pool; Figure 3).**

cross-encoders, and six zero-shot LLMs. The contribution is the benchmark and what it reveals. One design choice shapes the results: negatives are mined name-similar while successors are name-different, which targets the sub-problem left once a name-aware retriever has narrowed the field (*which name-similar candidate is the true successor?*) but also makes the pool exploitable. We therefore score every method on three candidate pools and judge it by its worst.

Contributions.

(1) **An open benchmark** (§3): 125,519 labeled pairs over 3.3M GLEIF entities, four hard-negative classes, held-out time, event-type, and jurisdiction splits, and a Wikidata marquee supplement.
(2) **A three-pool diagnostic** (§5.4, §5.6): the released pool, an unrelated miner's pool, and retrieval over the full universe. Methods that exploit the construction post the best name-input scores on the released pool and collapse on the other two.
(3) **Empirical findings** (§5): among name-only methods, a 22M-parameter cross-encoder is the one that transfers, 24–35 pp Hits@1 ahead of the best bi-encoder and ahead of six zero-shot LLMs on TIME and EVENT. The full structured record matters as much as architecture: a gradient-boosted matcher over it beats that cross-encoder on-pool by 18 pp, and a cross-encoder over it wins the fielded pool by 12 pp. Retrieval, not reranking, is the deployment bottleneck: Recall@20 of 33–50% caps end-to-end Hits@1 at 21–39%, echoing the blocking literature's focus on candidate generation [11].

2 Related Work

Classical entity resolution matches records by similarity over structured fields [4], and modern PLM cross-encoders on serialized records (Ditto [8], GraLMatch [3]) are now standard; the OpenSanctions Pairs corpus [10] benchmarks LLM matching at 755K pairs. All of this work adopts a *snapshot* framing rather than lifecycle-spanning matches. Closest in spirit is temporal record linkage [7], which pioneered matching drifting entities but predates modern encoders and open registries. AnyMatch [13] shows that small fine-tuned models rival LLM matchers at a fraction of the cost, and WDC Products [9] studies how hard-negative construction shapes benchmark difficulty, the axis §5.4 examines here. Finally, TempEL [12] time-stratifies entity linking against an evolving Wikipedia, but its task is mention-to-KB disambiguation rather than record-to-record matching over legal entities, and financial-filing knowledge graphs (FinReflectKG [1], from this conference) are built at a single time slice. None of them poses the lifecycle-bridge question directly.

3 The Sylvester Benchmark

3.1 Data sources

We construct Sylvester entirely from open, persistently-licensed sources:

- **GLEIF Level-1 LEI-CDF.** The LEI repository (3.3M entities, 2026-05-19 snapshot), CC0, refreshed every eight hours. Each `LEIRecord` carries legal name, jurisdiction, legal form, addresses, registration metadata, a dated `LegalEntityEvents` log, and a `SuccessorEntity` field linking predecessor to successor by LEI.
- **GLEIF Level-2 Relationship Records.** 473,839 typed inter-LEI relationships, used in hard-negative construction.
- **Wikidata.** High-profile M&A via SPARQL on P156 (*followed by*); a marquee supplement and cross-validation source.
- **SEC EDGAR.** `company_tickers.json` (10,354 records) gives a name↔CIK crosswalk to GLEIF for the U.S. public subset (3,334 confident links); released with the benchmark, not used in the evaluation.

What GLEIF Level-2 already provides. Level-2 records have six types: direct and ultimate accounting consolidation, fund management, sub-fund, feeder fund, and international branch. All describe who owns or manages whom; none records that one entity became another. In the same snapshot, 34 of our 31,824 positives (0.1%) have any Level-2 edge between predecessor and successor, 74 (0.2%) share a Level-2 parent, and only 0.6% of predecessors appear in Level-2 at all. Level-2 therefore supplies related-but-distinct *negatives* here, not positives. The raw positives are Level-1 `SuccessorEntity` links; Sylvester adds what they lack: resolution to registered successors, removal of administrative links, hard negatives, held-out splits, and alternative candidate pools.

3.2 Positive labels

Our gold-standard positives are GLEIF `SuccessorEntity` links: explicit predecessor-LEI → successor-LEI pairs maintained by GLEIF's network of Local Operating Units. The 2026-05-19 snapshot contains 50,202 records with a non-null `SuccessorEntity` element, 33,645 of which resolve to a `SuccessorLEI`. The rest point to successors that never registered an LEI, and we exclude them.

Label quality validation. A five-class auto-classifier on the successor pairs (using documented event types, normalized name

delta, jurisdiction, legal form) yields: `strong_real` (documented event + name change) 25,428; `likely_real_name_changed` 3,330; `likely_real_event_only` 1,704; `ambiguous` 1,355; **`likely_administrative`** 1,828 (5.4%, filtered out). We retain the first four classes (94.6%; `ambiguous` stays because every link is still LOU-curated), yielding 31,817 plausibly-real predecessors (31,824 positive rows; seven links are duplicated in the source export).

3.3 Hard negatives

For each positive predecessor we mine up to 80 hard negatives drawn from four classes:

- `parent_subsidiary` (221 pairs) — LEIs that share a Level-2 relationship-record edge (consolidation, fund-management, sub-fund) with the predecessor. These look related but are legally distinct.
- `same_name_same_country` (10,208) — distinct LEIs with identical normalized legal names in the same jurisdiction: coincidental name collision vs. true lifecycle linkage.
- `same_name_diff_country` (10,703) — distinct LEIs with identical normalized name in a different country.
- `token_overlap_same_country` (232,601) — fuzzy fallback: distinct LEIs in the same country sharing ≥ 2 content tokens after normalization, ranked by Jaccard. Covers the long tail when name-exact collisions are sparse.

After deduplication this leaves 253,733 candidate (predecessor, hard-negative) pairs covering 20,192 of 31,817 predecessors, or roughly 63%. The remaining 37% are small entities in jurisdictions without name collisions, and they fall back to random in-batch negatives during training.

3.4 Evaluation splits

For each positive we identify its *anchor lifecycle event* — the highest-priority COMPLETED event on the predecessor's record by type-priority then effective date. We then assign each positive (and its associated negatives) to one of three split regimes:

- **Held-out time:** anchor effective date $<$ 2025-01-01 (train) vs. $\geq$ 2025-01-01 (test); completed-event volume rises ≈4× in 2022 when GLEIF tightened reporting, so the effective window is 2022–2026. Test: 3,448 positives, 2,007 Hits@1-eligible anchors (§5.2).
- **Held-out event type:** train on M&A, rename, legal-form change; test on spinoff, demerger, breakup, absorption — dominated by *absorption* (2,609 of 2,760 anchors, 94.5%), the rest statistically thin (§5.5).
- **Held-out jurisdiction:** train on US + GB + DE; test on SG + JP + BR. Test composition: 161 anchors, 161 positives, of which 88 are Hits@1-eligible — the smallest split; its headline numbers carry ∼±8.5 pp 95% bootstrap CIs (vs. ∼±2.0–2.3 pp on time/event).

We refer to these regimes as TIME, EVENT, and JURIS. Each positive gets up to 5 hard negatives if its splits are all `train`, or up to 10 if any split places it in `test`, yielding 125,519 final pair rows.

Wikidata marquee supplement. Released alongside the benchmark: 302 high-profile M&A pairs from Wikidata P156 both-sides-resolved to GLEIF, 283 not captured by GLEIF `SuccessorEntity` (Sprint → T-Mobile, Wyeth → Pfizer, Interbrew → AB-InBev); not used in the headline evaluation.

4 Models

The neural models are standard sentence encoders, used off the shelf so that results speak to the benchmark rather than to a new architecture. A bi-encoder (§4.1) embeds each side independently and scores by cosine, which suits retrieval. A cross-encoder (§4.2) reads both records jointly, which suits reranking a short list and lets it compare specific tokens across records whose surface form has changed.

4.1 Bi-encoder

We fine-tune `intfloat/e5-small-v2` (33M params, 384-dim) via supervised contrastive learning with state features folded into the text prompt: "`query:` n_ℓ `|` `jurisdiction:` c_ℓ `|` `form:` f_ℓ". Training uses the 26,297 positives in the strict no-leakage intersection of all three split regimes (one hard neg per anchor), MultipleNegativesRanking loss (InfoNCE; in-batch + explicit hard negs), 3 epochs, batch 32, AdamW lr 2×10^{-5}, ≈6 min on Apple silicon. Ablation variants (in-batch-only; E5-cosine hard negs plus 1,372 transitive positives) are in Table 1. The bi-encoder uses contrastive loss and the cross-encoder BCE on the logit — different objectives by architectural convention.

4.2 Cross-encoder

We fine-tune a second-stage cross-encoder that ingests $(\text{record}_A, \text{record}_B)$ as a single concatenated sequence and emits a single relevance logit. The backbone is the 22M-parameter `ms-marco-MiniLM-L-6-v2` cross-encoder from HuggingFace; training uses the same positives as `method_v2_full` (26,297 plus 1,372 transitives) plus a 4:1 sample of E5-cosine hard negatives per positive, with positives labeled 1.0 and negatives 0.0. Loss: standard binary cross-entropy on the logit. 2 epochs, batch 32, lr 2×10^{-5}, 100 warmup steps. The extensions in §5.3 hold architecture/optimizer constant and modify only the serialized text, the training positives, and the backbone.

Inference cost. Scoring all 125,519 pairs takes ≈80 seconds on MPS; in deployment the cross-encoder re-ranks a retriever's top-K, and the benchmark's fixed 5–10-candidate pools make direct full scoring feasible (Table 1).

Hard-negative re-mining. Re-mining with E5-cosine top-20 same-country candidates, in place of the noisy token-overlap fallback, yields 652,764 negatives across 31,714 predecessors (vs. 253,733 over ≈20K before). The cleaner negatives are what the cross-encoder trains on, to large effect, and they are half of what later unlocks the full-record GBM's transfer (§5.4). For the bi-encoder they change nothing (§5.7).

5 Experiments

The analyses follow the challenges in §1: whether methods exploit the name-similar pool (§5.4, §5.6), how they handle skewed event types (§5.5), and whether they survive retrieval at registry scale (§5.6).

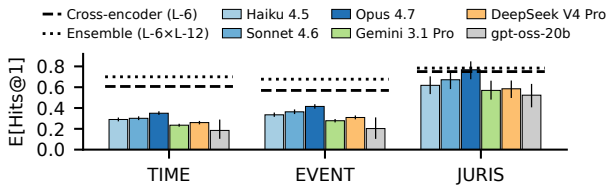


Figure 2: Zero-shot LLM panel: E[Hits@1] (95% CIs) for six models under the uniform single-pair protocol, against the fine-tuned L-6 (dashed) and ensemble (dotted; Table 1). Every LLM trails the L-6 on the two large splits; on JURIS (n=88) Opus meets it. `gpt-oss-20b`: 56/52/50-anchor sample.

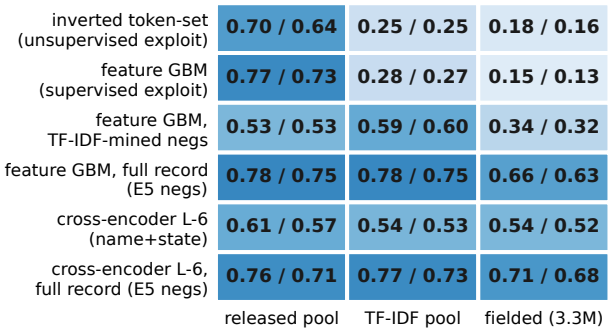


Figure 3: The argument in one exhibit: expected Hits@1 (TIME / EVENT; fill = mean) as the candidate pool shifts from the released pool to an unrelated miner's pool to retrieval from the full 3.3M-entity universe. Read rows by their *minimum*. The name-only exploits (rows 1–2) collapse off-pool; the name-feature GBM retrained on TF-IDF negatives (row 3) recovers only on its own miner's pool. Among name-input methods only the cross-encoder transfers (row 5). Given clean negatives *and* the full record, trees (row 4) and joint encoding (row 6) both transfer — and the full-record cross-encoder holds the highest minimum and wins the fielded pool (§5.4). The name+state TF-IDF cell re-trains that cross-encoder on E5-cosine negatives (`trn-E5`, §5.6), within 2 pp; fielded cells include the gold (strict conditional and end-to-end in the text); JURIS in the text and tables.

5.1 Baselines

We compare against four similarity baselines — `exact_name` (normalized name equality), `token_set` (token Jaccard), `tfidf_char` (char-3gram TF-IDF cosine), and `sentence_encoder` (zero-shot E5-small-v2, the method's backbone un-tuned) — and one *supervised* classical matcher, `features_gbm`: gradient-boosted trees over twelve string-similarity and field-agreement features in the Fellegi–Sunter lineage [4], trained on the released pool's no-leakage rows; the classical name-feature default (§5.4 upgrades it to the full record). Six zero-shot LLMs run under one protocol: one call per pair, byte-identical prompts carrying the same name+state inputs as the neural models (legal name, jurisdiction, legal form), a 0–10 integer rubric, `temperature=0` where exposed. Five API models from three organizations cover the full test set (26,564 pairs each) — **Claude Haiku 4.5**, **Sonnet 4.6**, **Opus 4.7**, **Gemini 3.1 Pro**, **DeepSeek V4 Pro** (thinking disabled) — and **gpt-oss-20b** (a fourth organization) runs locally on a 56/52/50-anchor sample (Table 1 footnotes).

Paired permutation tests on per-anchor expected Hits@1 separate the comparisons cleanly. The L-6 beats all five API models on TIME and EVENT at $p < 10^{-4}$ (+15 to +37 pp). The 88-anchor JURIS split is closer: the L-6 leads Haiku, Gemini, and DeepSeek ($p \leq 0.019$), sits within noise of Sonnet ($p = 0.14$), and is indistinguishable from Opus 4.7, which holds the higher point estimate (0.767 vs. 0.750, $p = 0.74$). The ensemble beats Opus on TIME/EVENT ($p < 10^{-4}$) and is a coin flip on JURIS. The panel (Figure 2) is strikingly vendor-heterogeneous: every Claude model outscores both other API vendors' frontier models, Haiku included, and §7 examines the elicitation-format confound behind part of that spread. Under Holm correction across the paper's ≈30 paired tests, every $p < 10^{-4}$ result survives; the $p \approx 0.02$ JURIS-level and on-pool-TIME margins do not, and should be read as directional.

5.2 Metrics

We report per-anchor Hits@1 and MRR alongside pairwise AUROC, and ship Hits@3, AUPRC, and threshold-tuned F1 with the released metrics. Discrete scorers tie often, so Hits@k and MRR are exact *expectations over uniformly random tie-breaking*; a fixed tie order is biased. Per-anchor metrics cover *eligible* anchors: a pool of at least two candidates with the gold present. The 37% of predecessors with no mined hard negative leave singleton pools and drop out (1,612 of 2,760 EVENT-test anchors are eligible; §3).

5.3 Headline results

Table 1 reports all methods on the released pool (`pairs.parquet`; token-overlap-mined negatives). The best bi-encoder lifts Hits@1 5–8 pp over zero-shot E5-small and the cross-encoder a further 24–35 pp. JURIS numbers (n=88) are directional throughout.

Extensions: enriched prompt + L-6×L-12 ensemble. Stacking prompt enrichment (GLEIF Level-2 parent/child names, founding year, ISO country, category), 259 Wikidata-mined training positives, multi-seed averaging, and an L-12 ensemble partner lifts Hits@1 +9 to +11 pp on time/event (+3 on jurisdiction, inside its CI; Table 2); baseline comparisons keep the single L-6 apples-to-apples. One negative result: symmetric scoring (averaging $s(a,b)$, $s(b,a)$) hurts −0.9/−1.9/−9.1 pp — consistent with a *directional* predecessor→successor representation (the TIME delta sits within its CI).

5.4 The construction cuts both ways: an anti-similarity baseline

`token_set`, `tfidf_char`, and `sentence_encoder` all fall below chance AUROC on time/event (Table 1). The cause is mechanical: hard negatives are deliberately name-similar to the predecessor while positives are by construction name-different, so on this

Table 1: Test-set metrics on the released pool. Bold = L-6 cross-encoder used in baseline comparisons. Random ranking scores 0.18–0.22 Hits@1 (pools average 7.4–8.3 candidates). Hits@1 95% CIs: ±2.0 / ±2.3 / ±8.5 pp. *`anti-token_set` (*negated* similarity) and the name-feature GBM exploit the construction and collapse off-pool; the two full-record models transfer across all three pools (Figure 3, §5.4). †Opus: no temperature parameter; Gemini (preview): thinking at lowest level, Batch API. ‡Local; 56/52/50-anchor sample.

	Held-out time			Held-out event			Held-out jurisdiction		
Method	AUROC	Hits@1	MRR	AUROC	Hits@1	MRR	AUROC	Hits@1	MRR
`exact_name`	0.474	0.163	0.373	0.523	0.193	0.400	0.571	0.352	0.539
`token_set`	0.193	0.136	0.319	0.266	0.180	0.356	0.323	0.356	0.522
`tfidf_char`	0.277	0.180	0.360	0.349	0.213	0.386	0.568	0.420	0.583
`sentence_encoder`	0.315	0.204	0.385	0.386	0.232	0.408	0.648	0.432	0.587
`anti-token_set` (1−Jaccard)	0.807	0.700	0.761	0.734	0.638	0.715	0.677	0.516	0.618
`features_gbm` (supervised)*	0.935	0.772	0.840	0.926	0.734	0.816	0.960	0.835	0.897
`features_gbm-full` (full record)*	0.929	0.783	0.860	0.912	0.753	0.838	0.893	0.881	0.924
LLM pairwise, zero-shot — uniform single-pair protocol, full test coverage									
`Claude Haiku 4.5`	—	0.290	0.476	—	0.335	0.517	—	0.618	0.738
`Claude Sonnet 4.6`	—	0.301	0.492	—	0.364	0.542	—	0.672	0.758
`Claude Opus 4.7`†	—	0.350	0.534	—	0.416	0.589	—	0.767	0.839
`Gemini 3.1 Pro`†	—	0.235	0.433	—	0.278	0.474	—	0.569	0.699
`DeepSeek V4 Pro`	—	0.260	0.451	—	0.309	0.497	—	0.585	0.712
`gpt-oss-20b`‡	—	0.185	0.377	—	0.203	0.403	—	0.523	0.683
`method_full` (bi-enc, token-overlap negs)	0.403	0.255	0.425	0.451	0.291	0.450	0.667	0.500	0.627
`method_no_hard` (bi-enc, no explicit negs)	0.407	0.259	0.429	0.461	0.299	0.459	0.668	0.511	0.641
`method_v2_full` (bi-enc, E5-cosine negs)	0.405	0.250	0.424	0.457	0.296	0.455	0.694	0.500	0.629
Cross-encoder (L-6, 22M)	**0.836**	**0.607**	**0.725**	**0.825**	**0.569**	**0.701**	**0.868**	**0.750**	**0.818**
Cross-encoder (L-12, 33M)	0.844	0.631	0.742	0.836	0.598	0.724	0.870	0.750	0.823
Cross-encoder (L-6, full record)*	0.930	0.761	0.845	0.915	0.708	0.808	0.954	0.875	0.916

Table 2: Cross-encoder extensions: Hits@1 [95% bootstrap CI]; L-6 row repeated from Table 1. Prompt enrichment is the largest increment on TIME/EVENT; the L-12 pairing captures the JURIS gain (CIs overlap — read per-step deltas as directional).

Model	TIME	EVENT	JURIS
Cross-encoder (L-6, 22M)	0.607 [0.588, 0.629]	0.569 [0.547, 0.593]	0.750 [0.659, 0.830]
+ enriched prompt	0.675 [0.653, 0.697]	0.642 [0.620, 0.662]	0.750 [0.659, 0.830]
+ Wikidata-LEI positives	0.694 [0.669, 0.712]	0.651 [0.630, 0.673]	0.773 [0.682, 0.853]
+ L-12 backbone (33M)	0.697 [0.678, 0.716]	0.676 [0.657, 0.697]	0.761 [0.670, 0.841]
Ensemble L-6×L-12	**0.700 [0.681, 0.718]**	**0.677 [0.656, 0.700]**	**0.784 [0.682, 0.864]**

candidate distribution name similarity is *evidence against* a true successor link. So we report the mirror image as its own baseline: `anti-token_set` ranks candidates by *negated* Jaccard, and on the curated pool it is strong (Table 1; *above* the single L-6 cross-encoder on time and event) because picking the least name-similar candidate exploits the construction.

The supervised `features_gbm` does the same thing, learned rather than hand-built. Given training labels it discovers the inversion along with the miners' residual signatures, posts the best name-only scores in the paper (Table 1; above the ensemble at $p < 10^{-4}$ on TIME/EVENT), and then collapses off-pool just as hard (Figure 3; on JURIS, 0.36 on Sylvester-TF-IDF and 0.12 fielded; AUROC across splits 0.54–0.57 and 0.45–0.52).

Is that collapse a fact about feature models, or about what we fed this one? It carries two handicaps we can remove, separately and together: the noisy v1 negatives and the name-only feature set. On the cross-encoder's own E5-mined negatives, the name-feature GBM improves cross-pool but still dies fielded (0.25 / 0.25 retrieval-conditional vs. 0.46 / 0.50); a TF-IDF-negative retrain behaves alike (Figure 3, row 3). Given the full structured record (addresses, previous names, liveness, registration status, dates, registrar; the successor fields the labels derive from are excluded) but v1 negatives, it wins the released pool (0.87 / 0.84) yet still fails to cross pools (0.38 / 0.35 on TF-IDF). Given *both*, it transfers across all three and wins (`features_gbm-full`): 0.78 / 0.75 on-pool (+18 pp over the L-6, $p < 10^{-4}$; +8 pp over the ensemble), 0.78 / 0.75 on TF-IDF, and 0.51–0.66 fielded retrieval-conditional, where the enriched ensemble leads by 1–3 pp on TIME/EVENT and trails by 6 pp on JURIS (all n.s.) — at negligible inference cost. Permutation importance credits fields no name-only method sees: candidate liveness, shared postal/city, previous-name overlap, same registrar. Nor is the transfer a live-vs-dead prior: ablating all four liveness/registration-status features costs 4–7 pp and transfer survives (0.72 / 0.72 on TF-IDF; 0.43 / 0.47 fielded-conditional, at parity with the L-6).

Completing the grid: a cross-encoder trained with the identical protocol but a serialized *full-record* prompt (the same fields the GBM sees) lands 2–5 pp behind the trees on-pool (0.76 / 0.71, $p \leq 0.018$), is within 2 pp on TF-IDF (0.77 / 0.73), and pulls clearly

Table 3: Per-event-type Hits@1 on the held-out-EVENT test split (Hits@1-eligible subset), zero-shot E5-small vs. L-6. The full test split is 94.5% ABSORPTION (2,609 of 2,760); SPINOFF at n=4 is statistically uninformative. Lift concentrates in the absorption-like types.

Event type	n anchors	zero-shot E5	Cross-enc (L-6)	Δ (pp)
ABSORPTION	1,546	0.230	**0.571**	+34.1
DEMERGER	42	0.333	**0.595**	+26.2
BREAKUP	20	0.150	**0.400**	+25.0
SPINOFF	4	0.500	0.250	−25.0*

* n=4, statistically uninformative.

ahead on the fielded pool: 0.64 / 0.66 / 0.78 retrieval-conditional, +12 pp over the GBM and +9–18 pp over the ensemble ($p < 10^{-4}$ on TIME/EVENT); best end-to-end here (21–39%).

The claims thus split by input class. Among name-input methods, the L-6 is the only one whose worst pool stays strong. Given the full record, trees and joint encoding tie on curated pools and the cross-encoder wins the fielded universe: much of the transferable signal is in fields no name-only method sees.

The unsupervised exploit collapses under every pool change we test: 0.25 / 0.25 / 0.20 Hits@1 on Sylvester-TF-IDF (chance 0.09) against 0.53–0.66 for the cross-encoders (Table 4), and 9–18% fielded. It is a one-way bet on acquirer-renaming: on TIME's M&A slice (n=1,506) it scores 0.741, above the L-6 (0.625) and the ensemble (0.711), though not the full-record models, but only 0.11 on pure renames and 0.35 on legal-form changes, where the cross-encoder scores 0.90 and 0.85. Curated-pool Hits@1 alone can therefore be gamed; within the name-input class, the cross-encoder's claim is being strong *simultaneously* on all three pools and on every event type with a usable sample. A plausible mechanism is joint attention: asking whether the *differences* between two records look like a lifecycle-event pattern, not how similar two strings are.

5.5 Per-event-type breakdown

Aggregate Hits@1 hides large variation across event types.

Held-out time. Pure renames (CHANGE_LEGAL_NAME, n=10) are the easiest slice for every similarity-direction method, zero-shot E5-small included (9 of 10, no lift); only the inverted exploit fails them (0.11, §5.4). The gains come instead from events where the post-event name is inherited from the acquirer and string similarity cannot bridge the gap: MERGERS_AND_ACQUISITIONS (n=1,506: 0.18 zero-shot → 0.62 L-6 → 0.71 ensemble; the inverted exploit's best slice, 0.74, §5.4) and ABSORPTION (n=461: 0.24 → 0.54 → 0.65). The ensemble also more than doubles zero-shot on the small but operationally critical BREAKUP subset (n=7: 0.14 → 0.71).

"Held-out event type" really means "held-out absorption." The held-out-event test set is 94.5% ABSORPTION anchors (§3.4); DEMERGER, BREAKUP, and SPINOFF contribute n=42, n=20, and n=4 within the Hits@1-eligible subset. Table 3 shows the lift is consistent across the absorption-like types and that the SPINOFF cell is statistically uninformative, so the aggregate "+34 pp Hits@1 over zero-shot E5" on this split is primarily a transfer claim about absorption, not a strong claim about spinoffs.

5.6 Robustness to candidate-pool shift

Both released miners (v1 token-overlap, v2 E5-cosine) rank by name-related signals, so we test whether the gap survives a pool built by a mechanically different miner.

Setup. We re-mine hard negatives with sparse char-3gram TF-IDF cosine, which has no learned representation and shares no parameters with E5, holding the other three classes, splits, and positives constant (Sylvester-TF-IDF). We train cross-encoders on the E5-cosine negatives (trn-E5) and the TF-IDF negatives (trn-TF-IDF) and score each on both pools. The released pool's negatives are token-overlap-mined, so the headline configuration is itself already cross-miner; the only miner-matched cell is trn-TF-IDF on the TF-IDF pool.

What the 2×2 tells us. The architecture generalizes across miners: each trained model lands within 1–2 pp of the other on the pool it never saw (TIME/EVENT), miner-matching is worth a bounded +7–11 pp, and the gain over zero-shot E5 survives every pairing (+13 to +40 pp, outside any CI band), though its size depends on the miner.

Fielded retrieve-then-rerank. A deployment pipeline retrieves top-K candidates from the full universe, then reranks — unlike the curated pools above. The trained method_v2_full bi-encoder retrieves top-20 per test anchor from the 3.3M-entity GLEIF universe, filtered to the anchor's jurisdiction (the lightest production filter); the cross-encoder reranks. The filter is nearly free (1.8–4.1% of positives cross countries, capping recall at ≥95.9%, far above the observed 33–50%): the bottleneck is the embedding, not the filter.

Bi-encoder Recall@20 is 33.2 / 39.6 / 49.7% on TIME / EVENT / JURIS. That is the hard ceiling for any reranker. When the bi-encoder does retrieve the true successor, the L-6 picks it #1 with 46–57% Hits@1, the ensemble 53–60%, the full-record GBM 51–66% (n.s. vs. the ensemble) — and the full-record cross-encoder 64–78%, ≥ 9 pp clear of every other reranker ($p < 10^{-4}$ on TIME/EVENT; §5.4). End-to-end deployment Hits@1 (recall × conditional, unretrieved truths counted as misses) is 15–28% for the L-6, 17–30% for the ensemble, 17–33% for the full-record GBM, and 21–39% for the full-record cross-encoder. The bottleneck is the one the anti-correlation analysis predicts: fixed embeddings retrieve by similarity, and lifecycle successors are often dissimilar.

5.7 Ablations

State features help modestly. Ablating jurisdiction and legal form from the prompt costs 1.1–1.8 pp Hits@1 on every split, most on held-out event type and least on held-out jurisdiction. Scale does not substitute: moving the bi-encoder backbone from E5-small (33M) to E5-base (110M) leaves TIME/EVENT unchanged (−0.2/ − 1.2 pp; JURIS +2.3), so on this step at least the ceiling looks structural rather than parametric.

Explicit hard negatives slightly hurt the bi-encoder. In Table 1, method_no_hard edges out the v1 full model by 0.4–1.1 pp Hits@1 on every split, and E5-cosine re-mining does not change that. The culprit is the token_overlap_same_country class, 92% of mined v1 negatives and the lowest-precision one; in-batch negatives from MultipleNegativesRanking are cleaner on their own. Better explicit

Table 4: Hits@1 with 95% bootstrap CIs for the 2×2 cross-encoder train/evaluation matrix; row labels name the training miner. The zero-shot E5 row is bootstrapped separately (within 0.002 of Table 1). The advantage over E5 holds in headline pairings (+20 to +40 pp) and swapped (+13 to +39 pp); re-trained `trn-E5` cells reproduce the original within 2 pp on time/event.

	Released pool			TF-IDF pool		
	TIME	EVENT	JURIS	TIME	EVENT	JURIS
E5-small zero-shot	0.204 ±1.8	0.232 ±2.1	0.432 ±10.5	0.265 ±1.4	0.320 ±1.7	0.441 ±7.8
cross-enc trn-E5	**0.601** ±2.2	**0.552** ±2.4	0.705 ±8.8	0.544 ±1.8	0.529 ±1.7	0.571 ±7.5
cross-enc trn-TF-IDF	0.589 ±2.0	0.534 ±2.4	**0.750** ±9.1	**0.655** ±1.5	**0.615** ±1.7	**0.640** ±7.1
Δ *vs. E5, headline pairing*	+39.7	+32.0	+27.3	+39.0	+29.5	+19.9
Δ *vs. E5, swapped pairing*	+38.5	+30.1	+31.8	+27.9	+20.9	+13.0

negatives, in other words, do not always help contrastive training, even though they are exactly what the cross-encoder needs (§4.2).

5.8 Additional diagnostics

Four diagnostics; full tables ship with the released code. **B-cubed cluster F1** [2] is 0.92 / 0.90 / 0.91 for the name+state cross-encoder and 0.94 / 0.93 / 0.94 for the full-record one, vs. ≤0.48 for any bi-encoder (each collapses to one giant cluster at the tuned threshold); restricted to true clusters of size ≥2, 0.85 / 0.84 / 0.88 (name+state). **Calibration and selective prediction.** ECE (10-bin) is 0.20 and Brier 0.18 for the name+state cross-encoder (0.13 / 0.12 full-record); temperature scaling does not fix the high-score over-confidence, so we report raw scores. A selective-prediction [5] sweep on the ensemble reaches Hits@1 0.93–0.98 at 20% coverage and 0.83–0.98 at 50%, so high-precision users can threshold. **Cross-domain transfer.** On zero-shot Magellan Walmart-Amazon [6] the cross-encoder *loses* to its own zero-shot base encoder by 5.9 pp AUROC: the representation is specialized to lifecycle legal-entity matching, not generic ER. **Label quality, second opinion.** A rule-based re-classifier, orthogonal to the primary's event-type evidence, agrees with it on 98.7% [95% CI 97.0–100%] of the 231/277 confidently-judged positives (46 uncertain), bounding detectable label noise there at ≈3%; human review is future work.

5.9 Where each method is fooled

Table 5 reports the mean score per negative class, where higher means more often fooled. Bi-encoders cut zero-shot E5's score on `parent_subsidiary` and `same_name_diff_country` by 40–52%, but `same_name_same_country` remains a wall at ≈0.89. The cross-encoder breaks it (0.89 → 0.43; 0.36 ensemble), and the ensemble nearly solves the other name classes (0.91 → 0.17 / 0.28). The one regression is `parent_subsidiary` (0.50 → 0.67 → 0.75 across bi-encoder, L-6, ensemble): the prompt enrichment that injects GLEIF L2 parent/child names makes it *worse*. At 99 held-out examples the headline impact is small, but the direction is clear: explicit multi-class modeling of corporate-hierarchy relationships.

Table 5: Negative-class mean score (lower is better) on negatives *never seen in training* (*n*=99 / 1,398 / 1,096 / 20,759): *p/s* parent/subsidiary, *sn-D* same-name/diff-country, *sn-S* same-name/same-country, *tok* token-overlap. The ensemble improves three classes but *worsens p/s* (0.67 → 0.75): prompt-injected L2 context makes parent/subsidiary pairs look *more* similar. Scores are on each method's own scale (compare down columns); bold = lowest among learned methods.

Method	p/s	sn-D	sn-S	tok
`tfidf_char`	0.220	0.590	0.783	0.595
`sentence_encoder`	0.841	0.912	0.960	0.905
`method_full`	0.501	0.438	0.893	0.747
Cross-encoder (L-6, 22M)	0.668	0.201	0.432	0.311
Ensemble (L-6×L-12)	0.748	**0.166**	**0.358**	**0.284**

6 Analysis

6.1 Error analysis

We sampled 100 `method_full` failures on held-out time: 66 mixed or other, 19 where anchor and true successor share *no* tokens, 8 substring-overlap distractors, and 7 exact normalized-name collisions. The no-token cases are hardest for name-input methods — they need world knowledge, graph context, or the full record (§5.4); Figure 1's anchor is representative: resolving "DB" to Deutsche Bank from name strings alone is beyond what a 33M encoder reliably does.

Residual JURIS failures cluster on sibling consolidation. Of the 19 / 88 anchors the ensemble mis-ranks on JURIS, at least 11 are sibling consolidations — the model returns a sibling vessel instead of the parent — plus Japanese-script renames the ASCII tokenizer cannot bridge. Multilingual pretraining and ownership-graph context would address both.

7 Limitations and Future Work

Headline numbers reflect the construction, not generic lifecycle ER.. Negatives are mined name-similar and positives are often name-different, so naive name similarity is anti-predictive on the released pool (§5.4). Headline numbers measure the sub-problem this construction defines and should not be read as generic

lifecycle-ER performance. The candidate distribution matters a great deal: scored against *random* same-country pools rather than mined ones, the marquee supplement (283 pairs, 244 anchors) inverts the picture. Token Jaccard reaches 0.41, the name+state cross-encoder falls below chance (0.07), and the full-record models recover only partly (0.21–0.24). Models trained on mined pools inherit their anti-similarity prior. Eligibility also conditions on a name-colliding neighbour existing (§5.2), which amplifies the effect.

Splits are pair-level, not entity-disjoint. 32% of TIME-test candidate entities and 21% of gold successors appear somewhere in training pairs (JURIS <1%). The L-6 shows no seen-entity advantage (unseen 0.62 vs. seen 0.57 on TIME, p=0.04 toward unseen; 0.57 vs. 0.58 on EVENT, n.s.), but an entity-disjoint split would be a cleaner test of generalization.

Deployment performance is modest. End to end, the strongest reranker reaches only 21–39% Hits@1, and the others 15–33%, because first-stage Recall@20 is 33–50% (§5.6). A stronger first stage (hybrid sparse–dense retrieval, or a pre-filtered shortlist) is the highest-leverage future work.

Generalization evidence is narrow. The held-out-event test set is 94.5% absorption, so EVENT results show transfer to absorption, not to spinoffs or breakups (§5.5). The jurisdiction test has 88 eligible anchors (±8.5 pp CI), so JURIS results are directional. GLEIF `SuccessorEntity` is filed voluntarily, which biases positives toward well-counseled regulated entities; 283 of 302 marquee Wikidata M&A pairs are absent from it. A second-classifier check bounds detectable label noise at ≈3% (§5.8); human review is future work.

LLM baselines bound one elicitation format. On TIME/EVENT the 0–10 rubric costs the five API models 13–48 pp Hits@1 in within-pool ties: Gemini 3.1 Pro scores every candidate identically on 55% of TIME anchors, and oracle tie-breaking would lift it from 0.23 to 0.72, whereas Sonnet and Opus tie only ≈5% of pools. Batched Sonnet 4.6 also beats single-pair on the same 56 TIME anchors (0.46 vs. 0.39). The panel bounds what this format extracts, not what these models can do; finer scales, log-probs, or a full-record prompt would likely raise it, and cross-vendor gaps may partly reflect exposure to GLEIF-derived text.

Smaller caveats. Over three seeds the bi-encoder and enriched cross-encoder vary ≤1.6 pp and the full-record models ≤3.4 pp on-pool; fielded, the full-record cross-encoder's worst seed still leads the GBM's best by 8 pp on TIME/EVENT. Of the hard negatives, 0.22% appear in GLEIF's `SuccessorEntity`; dropping them moves results ≤0.2 pp. Parent/subsidiary negatives regress under prompt enrichment (§5.9); at 99 examples the effect is weak, but related-but-distinct entities merit explicit modeling.

8 Conclusion

Sylvester is an open benchmark for lifecycle-aware entity resolution on financial entities, built from public GLEIF, SEC EDGAR, and Wikidata data, with a three-pool protocol that separates methods that learn the pool from methods that learn the task. Name-only exploits post the best name-input scores and then collapse off-pool; among name-input methods, a 22M-parameter cross-encoder survives and stays ahead of every zero-shot LLM we test on the two large splits. Input matters as much as architecture: boosted trees over the full registry record transfer across all three pools, and a cross-encoder over the same record wins the fielded universe by 12 pp. The evidence is strongest for held-out time; event-type transfer is mostly transfer to absorption, and jurisdiction results are directional. What remains open is the first stage: with 33–50% retrieval recall, it is candidate generation, not matching, where lifecycle ER breaks.

Reproducibility. Code (MIT), data (CC-BY-4.0), parsers, predictions, trained weights, and a datasheet are at https://github.com/chang-07/er-research. Every non-LLM result reproduces from raw GLEIF in ≈2 hours on one MacBook.

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